EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

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Aim

To implement and explore the security, access-control, data-sharing and governance features of a Zoho Creator application within the context of a **Gym Membership Management System**.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the **Gym Membership Management System** application. The application captures prospective and active member registrations through the `Gym_Member_Form` (or `Pre - Gym member form`), which stores personal health metrics, contact information, subscription plans and payment consent. Because these records contain personal health and financial data of fitness members, the application served as a suitable practical environment for exploring the security and governance features described in this record.

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Gym Membership Management application. It illustrates how internal staff and external members interact with system boundaries.

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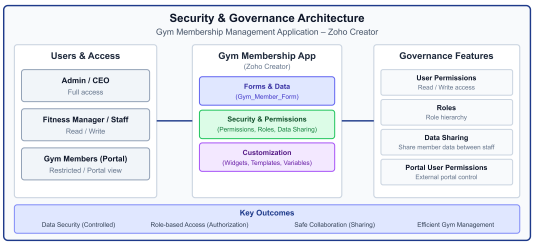


Figure 1: Security & Governance Architecture (Conceptual Diagram)

The diagram shows users (such as the Gym CEO / Administrator, Fitness Managers, Trainers with Read/Write access, and external portal users such as members) accessing the application, which contains member forms and data, security and permissions (user permissions, roles, and data sharing) and customisation items. The governance features manage and control access to member profiles and health records.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles, and Data Sharing, and is used to add and manage the permissions and roles of gym management staff and define record visibility. The Permissions tab lists predefined permission sets:

Name	Description (as displayed)
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components

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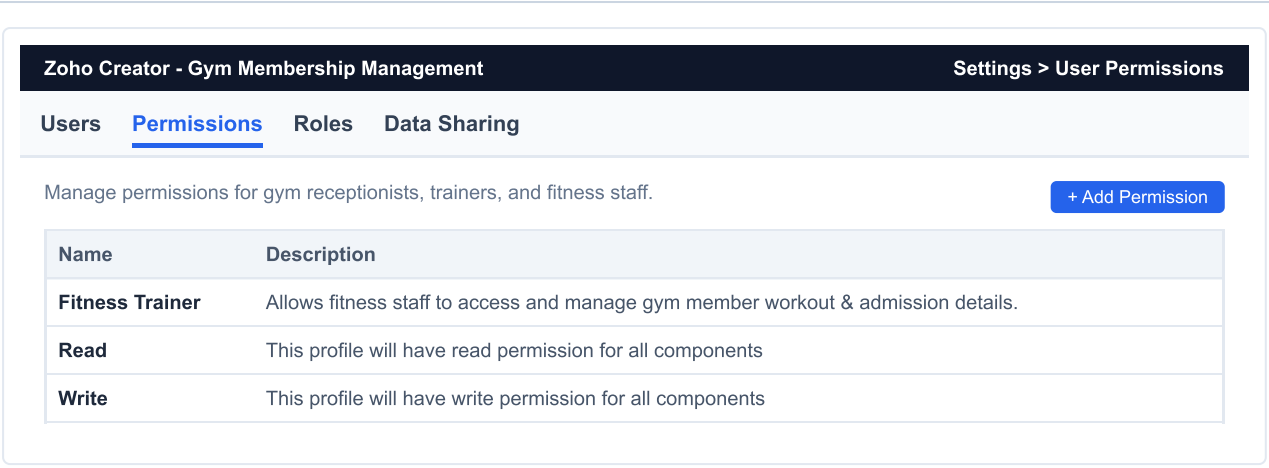


Figure 2: User Permissions in the Gym Membership Management application

Read permission allows a trainer or desk clerk to view member registration logs and attendance without modifying tier settings, whereas Write permission permits fitness staff to make changes, such as modifying membership durations, upgrading plans, or renewing subscriptions.

Role-Based Access Control

The Roles tab displays the organizational hierarchy of the gym facility. In the application, the root role CEO / General Manager is displayed, and options like Expand All, Collapse All, Manage Role Hierarchy, and Add Role are available.

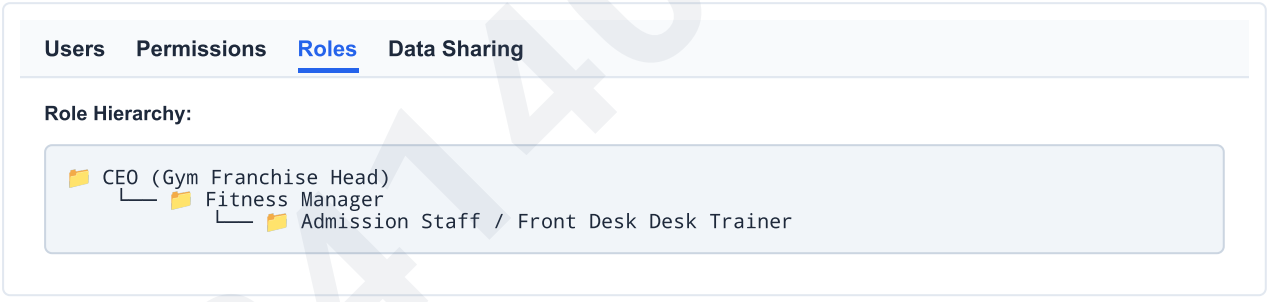


Figure 3: Roles and Role Hierarchy in the Gym Membership Management application

The Add Role option was used to configure a custom role for gym operational staff with the following parameters:

Field	Value
Role Name	Fitness Manager
Reports To	CEO
Description	Manages gym registrations, member health profiles, and trainer schedules
Share Data with Peers	Not selected

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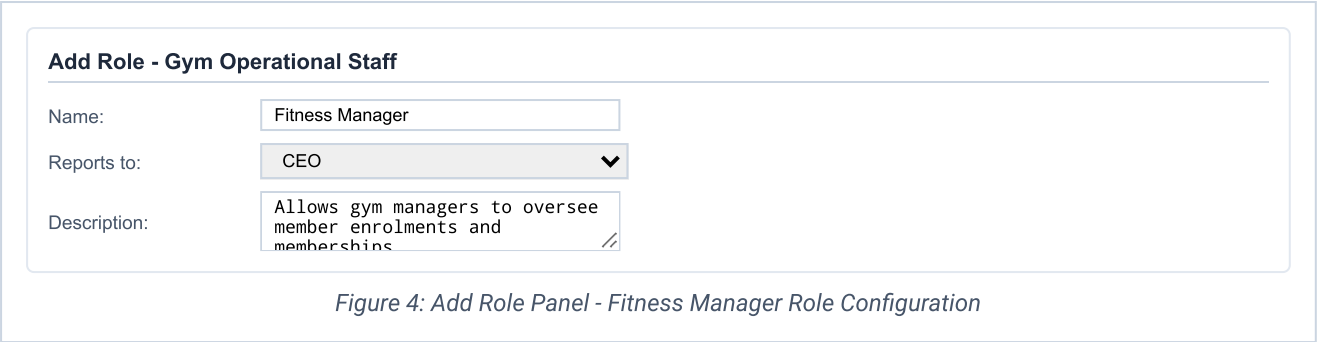


Figure 4: Add Role Panel - Fitness Manager Role Configuration

The Fitness Manager role was configured so that front-office gym trainers and desk staff report directly to the CEO/Owner in the operational hierarchy. This ensures proper administrative delegation for gym record auditing.

Data Sharing

Data sharing rules control the visibility of gym membership form records. A rule specifies the form, the role or user from which records are shared, the recipient role/user, and the designated access level. A data sharing rule was configured as follows:

Parameter	Value
Rule Name	Gym Member Data Sharing
Form	Gym_Member_Form (Pre - Gym member form)
Shared From	CEO (role)
Shared To	JASHWANTH SUNDARAJAN (user)
Access Level	Read/Write



Figure 5: Data Sharing rule configured for the Gym Member Form

The saved rule appears in the Data Sharing overview list. With Read/Write access, user **JASHWANTH SUNDARAJAN** can both view and modify the shared gym member records belonging to the CEO role.

Security Features and Governance Controls

Figure 6 is a conceptual academic diagram that groups the security and governance controls of Zoho Creator for managing membership records.

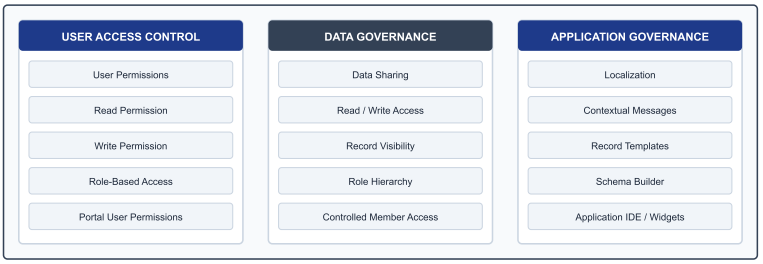


Figure 6: Security Features and Governance Controls in Zoho Creator (Conceptual Diagram)

Control Category	Explanation
User Access Control	User permissions (Read and Write), role-based access and portal user permissions determine who can access member files and what operations they can perform.
Data Governance	Data sharing rules, read/write access levels, record visibility and role hierarchy govern how gym registration data is accessed across branches.
Administrative Controls	Allow the gym system administrator to manage permissions, staff roles, registration templates, and application settings.

In this practical, user permissions and staff roles were evaluated, and a data sharing rule was configured for gym records.

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Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. For this application, such users are gym members who log in to view their membership validity, trainer schedules, and payment receipts.

Role-Based Access and Data Sharing Flow

Figure 7 is a conceptual diagram of the security flow: User → User Permissions → Role → Data Sharing Rule → Access Level → Gym Member Data.



Figure 7: Role-Based Access and Data Sharing Flow (Conceptual Diagram)

- A user is first given permissions (Read/Write).
- The user belongs to an operational role, such as CEO or Fitness Manager, which determines their position in the facility hierarchy.
- A data sharing rule specifies gym records shared from one role/user to another.
- The access level defines what operations the recipient can perform on gym member data.

Governance Observations

- Predefined Read and Write permissions were available in the application.
- Roles could be accessed, and a custom role configuration for **Fitness Manager** was prepared.
- Data Sharing was configured with Read/Write access for `Pre - Gym member form`.

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The practical successfully demonstrated how access to sensitive gym applicant and active member data can be controlled through permissions, roles, and data sharing rules in a low-code platform environment.

Conclusion

Zoho Creator provides effective built-in mechanisms for managing users, roles, permissions, and data visibility. In this experiment, the predefined Read and Write permissions were examined, the role hierarchy with the default CEO role was viewed, and a data sharing rule was configured to share the records of the **Gym Member Form** (Pre - Gym member form) from the CEO role to the user **JASHWANTH SUNDARAJAN** (Register No: 241401032) with Read/Write access.

The experiment highlights the essential security tools Zoho Creator offers for real-world low-code business applications. Controlling who can access gym membership data, and at what privilege level, remains a critical requirement for preserving member privacy, securing fitness health data, and ensuring reliable administrative operations.

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